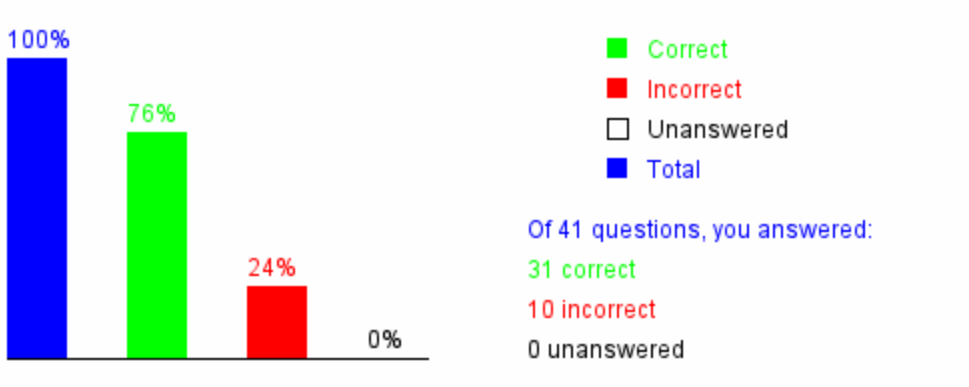
Introduction to Programming Using Python, Y. Daniel Liang

Many questions in this edition have been updated in the new edition. Please check with the publisher on the newest edition.

This quiz is for students to practice. A large number of additional quiz is available for instructors using Quiz Generator from the Instructor's Resource Website. Videos for Java, Python, and C++ can be found at https://yongdanielliang.github.io/revelvideos.html.

Chapter 14 Tuples, Sets, and Dictionaries



Please send suggestions and errata to Dr. Liang at y.daniel.liang@gmail.com. Indicate which book and edition you are using. Thanks!

Section 14.2 Tuples

14.1 Suppose t = (1, 2, 4, 3), which of the following is incorrect?

- A. `print(t[3])`
- B. `t[3] = 45`
- C. `print(max(t))`
- D. `print(len(t))`
- E. `print(sum(t))`

Your answer is correct

14.2 Suppose t = (1, 2, 4, 3), t[1 : 3] is _____.

- A. `(1, 2)`
- B. `(1, 2, 4)`
- C. `(2, 4)`
- D. `(2, 4, 3)`
- E. `(1, 2, 4, 3)`

Your answer is correct

14.3 Suppose t = (1, 2, 4, 3), t[1 : -1] is _____.

- A. `(1, 2)`
- B. `(1, 2, 4)`
- C. `(2, 4)`
- D. `(2, 4, 3)`
- E. `(1, 2, 4, 3)`

Your answer is correct

14.4 Suppose t = (1, 2, 4, 3, 8, 9), [t[i] for i in range(0, len(t), 2)] is _____.

A. `[2, 3, 9]`
B. `[1, 2, 4, 3, 8, 9]`
C. `[1, 4, 8]`
D. `(1, 4, 8)`
E. `(2, 3, 9)`

Your answer is correct

14.5 Suppose t = (1, 2), 2 * t is _____.

A. `(1, 2, 1, 2)`
B. `[1, 2, 1, 2]`
C. `(1, 1, 2, 2)`
D. `[1, 1, 2, 2]`
E. `illegal`

Your answer is correct

14.6 Suppose t1 = (1, 2, 4, 3) and t2 = (1, 2, 3, 4), t1 < t2 is _____.

A. `True`
B. `False`

Your answer is correct

Section 14.3 Sets

14.7 Which of the following statements produces {'a', 'b', 'c'}?

A. `list("abac")`
B. `tuple("abac")`
C. `set("abac")`
D. `None`

Your answer is correct

14.8 You can use _____ to create an empty set.

A. `{ }`
B. `( )`
C. `[ ]`
D. `set()`

Your answer is correct

14.9 Given two sets s1 and s2, s1 < s2 is _____.

A. `true if len(s1) is less than len(s2)`
B. `true ifthe elements in s1 are compared less than the elements in s2.`
C. `true if s1 is a proper subset of s2`
D. `true if s1 is a proper superset of s2`
E. `illegal`

Your answer B is incorrect
Click here to show the correct answer

14.10 Suppose s = {1, 2, 4, 3}, _____ returns 4.

- [ ] A. `sum(s)`
- [x] B. `len(s)`
- [ ] C. `min(s)`
- [x] D. `max(s)`
- [ ] E. `None`

Your answer is correct

14.11 Suppose s = {1, 2, 4, 3}, which of the following is incorrect?

- [x] A. `print(s[3])`
- [x] B. `s[3] = 45`
- [ ] C. `print(max(s))`
- [ ] D. `print(len(s))`
- [ ] E. `print(sum(s))`

Your answer is correct

14.12 Suppose s = {1, 2, 4, 3}, what happens when invoking s.add(4)?

- ( ) A. `There is no add method for a set object.`
- ( ) B. `This method is executed fine and 4 is added to the set.`
- ( ) C. `Since 4 is already in the set, Python raises a KeyError exception.`
- ( ) D. `You cannot add an element from a set.`
- (•) E. `This method is executed fine and 4 is not added to the set since 4 is already in the set.`

Your answer is correct

14.13 Suppose s = {1, 2, 4, 3}, what happens when invoking s.remove(12)?

- ( ) A. `There is no remove method for a set object.`
- ( ) B. `This method is executed fine and no exception is raised.`
- ( ) C. `Since 12 is not in the set, Python raises a KeyError exception.`
- (•) D. `You cannot remove an element from a set.`

Your answer D is incorrect
Click here to show the correct answer

14.14 Suppose s1 = {1, 2, 4, 3} and s2 = {1, 3, 4, 2}, _____ is true.

- (•) A. `s1 == s2`
- ( ) B. `s1 != s2`

Your answer is correct

14.15 Suppose s1 = {1, 2, 4, 3} and s2 = {0, 1, 5, 3, 4, 2, 13}, _____ is true.

- [x] A. `s1.issubset(s2)`
- [ ] B. `s1.issuperset(s2)`
- [ ] C. `s2.issubset(s1)`
- [x] D. `s2.issuperset(s1)`

Your answer is correct

14.16 Suppose s1 = {1, 2, 4, 3} and s2 = {1, 5, 4, 13}, what is s1 | s2?

- A. `{1, 2, 4, 3, 1, 5, 4, 13}`
- B. `{1, 2, 4, 3, 5, 13}`
- C. `{1, 2, 4, 3}`
- D. `{1, 5, 4, 13}`

Your answer is correct

14.17 Suppose s1 = {1, 2, 4, 3} and s2 = {1, 5, 4, 13}, what is s1 - s2?

- A. `{2, 3, 5, 13}`
- B. `{1, 2, 4, 3, 5, 13}`
- C. `{1, 2, 4, 3}`
- D. `{1, 5, 4, 13}`
- E. `{2, 3}`

Your answer is correct

14.18 Suppose s1 = {1, 2, 4, 3} and s2 = {1, 5, 4, 13}, what is s1 & s2?

- A. `{2, 3, 5, 13}`
- B. `{1, 2, 4, 3, 5, 13}`
- C. `{1, 4}`
- D. `{1, 5, 4, 13}`
- E. `{2, 3}`

Your answer is correct

14.19 Suppose s1 = {1, 2, 4, 3} and s2 = {1, 5, 4, 13}, what is s1 ^ s2?

- A. `{2, 3, 5, 13}`
- B. `{4, 3, 5, 13}`
- C. `{1, 4}`
- D. `{1, 5, 4, 13}`
- E. `{2, 3}`

Your answer E is incorrect
Click here to show the correct answer

14.20 Which of the following statement is false?

- A. `Lists are mutable`
- B. `Tuples are mutable`
- C. `Sets are mutable`
- D. `Strings are mutable`

Your answer is correct

14.21 You can have duplicate elements in a _____?

- A. `list`

- [x] B. `tuple`
- [ ] C. `set`

Your answer is correct 

14.22 The elements in a _____ are ordered?

- [ ] A. `list`
- [x] B. `tuple`
- [x] C. `set`

Your answer BC is incorrect 
Click here to show the correct answer

14.23 Suppose s = {1, 2}, 2 * s is _____.

- (•) A. `(1, 2, 1, 2)`
- ( ) B. `[1, 2, 1, 2]`
- ( ) C. `(1, 1, 2, 2)`
- ( ) D. `[1, 1, 2, 2]`
- ( ) E. `illegal`

Your answer A is incorrect 
Click here to show the correct answer

14.24 Suppose s1 = {1, 2, 4, 3} and s2 = {1, 5, 4, 13}, s1 + s2 is _____?

- ( ) A. `{1, 2, 4, 3, 1, 5, 4, 13}`
- (•) B. `{1, 2, 4, 3, 5, 13}`
- ( ) C. `{1, 2, 4, 3}`
- ( ) D. `{1, 5, 4, 13}`
- ( ) E. `illegal`

Your answer B is incorrect 
Click here to show the correct answer

Section 14.5 Dictionary

14.25 Which of the following statements create a dictionary?

- [x] A. `d = {}`
- [x] B. `d = {"john":40, "peter":45}`
- [x] C. `d = {40:"john", 45:"peter"}`
- [ ] D. `d = (40:"john", 45:"peter")`

Your answer is correct 

14.26 Suppose d = {"john":40, "peter":45}, the keys are _____

- ( ) A. `"john", 40, 45, and "peter"`
- (•) B. `"john" and "peter"`
- ( ) C. `40 and 45`
- ( ) D. `d = (40:"john", 45:"peter")`

Your answer is correct

14.27 Suppose d = {"john":40, "peter":45}, "john" in d is _____

- (●) A. `True`
- ( ) B. `False`

Your answer is correct

14.28 Suppose d1 = {"john":40, "peter":45} and d2 = {"john":466, "peter":45}, d1 == d2 is _____.

- ( ) A. `True`
- (●) B. `False`
- ( ) C. `illegal`

Your answer is correct

14.29 Suppose d1 = {"john":40, "peter":45} and d2 = {"john":466, "peter":45}, d1 > d2 is _____.

- ( ) A. `True`
- (●) B. `False`
- ( ) C. `illegal`

Your answer B is incorrect
Click here to show the correct answer

14.30 Suppose d = {"john":40, "peter":45}, d["john"] is _____

- (●) A. `40`
- ( ) B. `45`
- ( ) C. `"john"`
- ( ) D. `"peter"`

Your answer is correct

14.31 Suppose d = {"john":40, "peter":45}, to delete the entry for "john":40, use _____.

- ( ) A. `d.delete("john":40)`
- ( ) B. `d.delete("john")`
- (●) C. `del d["john"]`
- ( ) D. `del d("john":40)`

Your answer is correct

14.32 Suppose d = {"john":40, "peter":45}, to obtain the number of entries in dictionary, use _____.

- (●) A. `d.size()`
- ( ) B. `len(d)`
- ( ) C. `size(d)`
- ( ) D. `d.len()`

Your answer A is incorrect
Click here to show the correct answer

14.33 What will be displayed by the following code?

```
d = {"john":40, "peter":45}
print(list(d.keys()))
```

- (●) A. `["john", "peter"]`
- ( ) B. `["john":40, "peter":45]`
- ( ) C. `("john", "peter")`
- ( ) D. `("john":40, "peter":45)`

Your answer is correct

14.34 Suppose d = {"john":40, "peter":45}, what happens when retieving a value using d["susan"]?

- ( ) A. `Since "susan" is not a value in the set, Python raises a KeyError exception.`
- ( ) B. `It is executed fine and no exception is raised, and it returns None.`
- (●) C. `Since "susan" is not a key in the set, Python raises a KeyError exception.`
- ( ) D. `Since "susan" is not a key in the set, Python raises a syntax error.`

Your answer is correct

14.35 Suppose d = {"john":40, "peter":45}, what happens when retieving a value using d.get("susan")?

- ( ) A. `Since "susan" is not a value in the set, Python raises a KeyError exception.`
- (●) B. `It is executed fine and no exception is raised, and it returns None.`
- ( ) C. `Since "susan" is not a key in the set, Python raises a KeyError exception.`
- ( ) D. `Since "susan" is not a key in the set, Python raises a syntax error.`

Your answer is correct

14.36 Which of the following statements are true?

- ( ) A. `A Python list is immutable if every element in the list is immutable.`
- ( ) B. `A Python set is immutable if every element in the set is immutable.`
- (●) C. `A Python tuple is immutable if every element in the tuple is immutable.`
- ( ) D. `A Python tuple is immutable.`

Your answer is correct

14.37 Which of the following is a Python list?

- (●) A. `[1, 2, 3]`
- ( ) B. `(1, 2, 3)`
- ( ) C. `{1, 2, 3}`
- ( ) D. `{}`

Your answer is correct

14.38 Which of the following is a Python tuple?

- ( ) A. `[1, 2, 3]`
- (●) B. `(1, 2, 3)`
- ( ) C. `{1, 2, 3}`

D. {}

Your answer is correct

14.39 Which of the following is a Python set?

A. [1, 2, 3]
B. (1, 2, 3)
C. {1, 2, 3}
D. {}

Your answer B is incorrect
Click here to show the correct answer

14.40 Which of the following is a Python dictionary?

A. [1, 2, 3]
B. (1, 2, 3)
C. {1, 2, 3}
D. {}

Your answer C is incorrect
Click here to show the correct answer

14.41 Which of the following sets is equal to {1, 2, 3}?

A. {1, 2, 3}
B. {2, 1, 3}
C. {3, 2, 1}
D. {2, 3, 1}
E. {1, 2, 3, 2, 1, 3}

Your answer is correct